

Confluent Kafka Isotope

`confluent-kafka-isotope` is a reference implementation of an **e-commerce order pipeline that uses Kafka Interceptors, Prometheus, and Apache Flink to capture, analyze, and report end-to-end event-tracing data in both batch and near real time.**

Much like isotopes used to trace molecules through a biochemical pathway, each event carries lightweight metadata that allows it to be followed as it moves through Kafka topics and distributed microservices.

Kafka topics become the connective tissue between services, while Kafka Interceptors quietly transform the event pipeline itself into an observable distributed system.

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1.0 How an Isotope Traverses an Event Pipeline

An **Isotope** is a **lightweight tracing artifact attached to Kafka record headers**. Like a biochemical isotope used to trace molecules through a metabolic pathway, it allows the journey of a record through an event-driven architecture to be observed and analyzed.

This project models a Kafka pipeline as a **bipartite graph**: *services occupy one vertex set, topics the other, and every produce and consume operation forms an edge between them*. The resulting graph provides a unified view of the complete event topology, from producers to intermediate processing services to terminal consumers.

A `ProducerInterceptor` stamps each record with an isotope and appends one hop for every `send()` operation, capturing the produce edges. Consumers call `IsotopeContext.recordConsume(...)` to emit a lightweight marker representing the corresponding consume edges, while services that consume and then produce invoke `IsotopeContext.adoptFromRecord(...)` so the trace identity persists across every hop. Apache Flink reconstructs the complete **service → topic → service** graph from the isotope headers alone, producing reports such as end-to-end latency, topology discovery, coverage analysis, duplicate and loss detection, and forensic trace replay. The same implementation runs

unchanged on both Confluent Platform (self-managed) with Confluent Managed Flink (CMF) and Confluent Cloud for Apache Flink (CCAF).

Full design — the header layout (`x-isotope` JSON + seven scalar headers with a worked example), how the producer interceptor gets invoked, why the consume side uses explicit calls instead of a `ConsumerInterceptor`, and the bipartite-graph rationale — is in [docs/design.md](#).

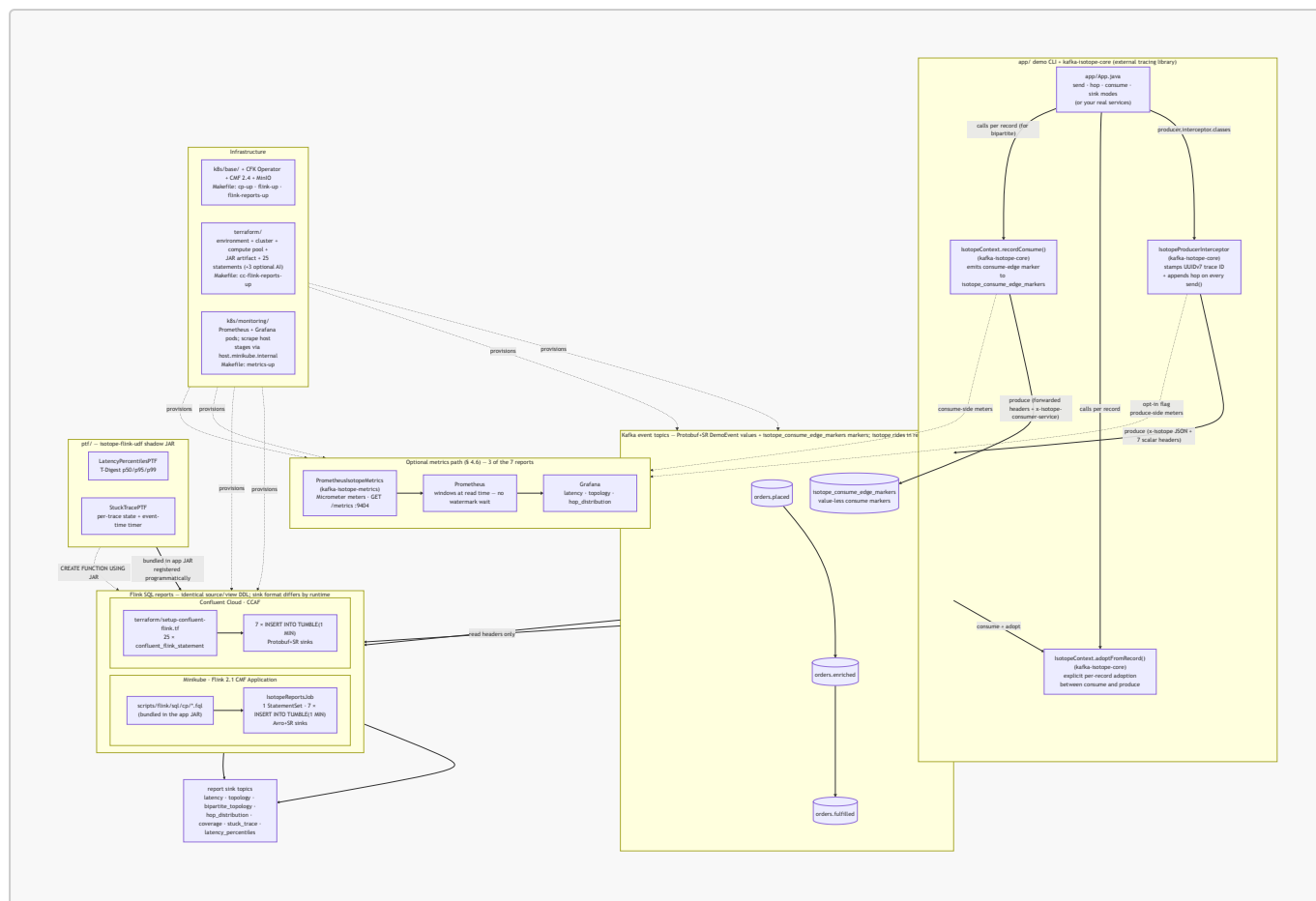
2.0 Architecture

A bird's-eye view of the moving parts. The demo CLI in `app/` consumes the external tracing library (`ai.signalroom:kafka-isotope-core`), which registers a Kafka `ProducerInterceptor` that stamps an isotope into record headers on every `send()`. Consume-then-produce services propagate the inbound trace by explicitly calling `IsotopeContext.adoptFromRecord(record)`. Business events then flow through a three-topic Kafka pipeline, where Flink SQL reads the isotope metadata and emits one-minute aggregate reports.

The same source and view DDL deploy unchanged to both runtimes. **Confluent Platform (CP)** on Minikube executes the `.fql` files under `scripts/flink/sql/cp/` as a single Flink 2.1 CMF Application (`IsotopeReportsJob`), while **Confluent Cloud for Apache Flink (CCAF)** applies the same logical SQL as inline `confluent_flink_statement` Terraform resources under `terraform/`. The shadow JAR from `ptf/`—which powers two of the seven reports—runs unchanged on both runtimes: bundled into the CP application JAR and uploaded as a Flink artifact on CCAF.

Alongside that—**additive, opt-in, and disabled by default**—the interceptor can also emit **Micrometer** metrics for **Prometheus**, with **Grafana** providing visualization ([§ 4.6](#)). This path produces the three **stateless** reports (`latency_1m`, `topology_1m`, and `hop_distribution_1m`) without requiring a stream processor. The remaining four reports continue to run in Flink because they depend on per-`trace_id` state or absence-of-event analysis, which falls outside Prometheus's query model.

(Kafka is drawn once below for brevity; each runtime provisions its own Kafka cluster.)



See § 3.0 for the file tree behind each box, and § 4.0 for the run commands.

3.0 Project layout

The isotope tracing library lives in its own repo — [j3-signalroom/kafka-isotope](#) ([ai.signalroom:kafka-isotope-core](#) + [ai.signalroom:kafka-isotope-metrics](#)). This repo is the runnable demo that consumes it.

app/ isotope library)	demo CLI + tests (consumes the
src/main/proto/ai/signalroom/kafka/isotope/proto/ demo_event.proto	DemoEvent message (Protobuf value
schema)	
src/main/java/ai/signalroom/kafka/isotope/ App.java	demo CLI – pipeline-position verbs
	(place / enrich / fulfill / ship) +
generic	send / hop / consume / sink modes;
registers	IsotopeProducerInterceptor + starts
the	PrometheusIsotopeMetrics exporter
(§ 4.6)	
src/integrationTest/java/.../ ProducerInterceptorIT,	BrokerSmokeIT,
BipartiteTopologyIT,	ThreeStageHopPropagationIT,

tests; produce/consume	IsotopeTestHarness – live-broker
	DemoEvent via SR-framed Protobuf (need Minikube CP + SR port-
forwarded)	
ptf/	Flink PTF shadow JAR (powers 2 of 7
reports)	
src/main/java/ai/signalroom/kafka/isotope/flink/	
LatencyPercentilesPTF.java	T-Digest p50/p95/p99 (PTF: per-
window state + timers)	
StuckTracePTF.java	per-trace state + event-time timer
TDigests.java	shared T-Digest (de)serialization
src/test/java/.../	TDigestsTest
k8s/base/	CFK manifests (applied by `make cp-
up` / `flink-up`)	
confluent-platform-c3++.yaml	Kafka / SR / Connect / ksqlDB /
Control Center	
flink-cluster-deployment.yaml	optional cp-flink session cluster
for ad-hoc	
sql`	`make flink-deploy` / `make flink-
flink-rbac.yaml	
k8s/monitoring/	RBAC for the cp-flink operator
`make metrics-up`	optional metrics showcase (§ 4.6) –
00-namespace.yaml	dedicated 'monitoring' namespace
10-prometheus.yaml	Prometheus pod/Service; scrapes
host stages	
	via
host.minikube.internal:9410/9411/9412	
20-grafana.yaml	Grafana pod/Service; auto-
provisioned datasource	
	+ 8-panel dashboard for the 6
produce/consume meters	
kustomization.yaml	`kubectl apply -k k8s/monitoring`
README.md	runbook + troubleshooting
scripts/	
port-forward-kafka.sh	localhost:30092 → Kafka,
localhost:8081 → SR	
port-forward-taskmanager.sh	Flink TaskManager web UI forward
deploy-cmf-flink-reports.sh	builds shadow JAR + uploads it as a
cmf://	
	artifact + deploys the reports as a
Flink 2.1	
	CMF Application (runs sql/cp/*.fql)
deploy-cc-flink-reports.sh	builds shadow JAR + wraps
`terraform apply`	
	for the CCAF path
cc-cli-env.sh	pulls Kafka + SR creds from
`terraform output`,	
	builds the JAAS string, exports
BOOTSTRAP /	
	SR_URL / KAFKA_KEY / KAFKA_SECRET /
JAAS / ...	
cc-app-run.sh	thin wrapper around `./gradlew

```

:app:run` that

the six -D flags
  flink/README.md
(CP=7 reports/Avro+SR,

layout, operations
  flink/sql/cp/
01_register_functions,

06_consume_events_view,

reports, 99_teardown

terraform/setup-confluent-flink.tf.)
  flink/sql/ccaf-ai/
off by default)
  trace_rca.fql
turns each

cause

ML_PREDICT

ai.tf)
terraform/
cc-flink-reports-up`)
  providers.tf
key/secret vars
  versions.tf
provider versions
  variables.tf
region, day_count
  data.tf
sources
  setup-confluent-environment.tf
governance package)
  setup-confluent-kafka.tf
rotation module

tf_module)
  setup-confluent-flink.tf
compute pool,

rotation, and 25 inline

resources: 4 ALTER TABLE

DROP FUNCTION +

INSERT INTO
  setup-ccaf-ai.tf
extra

```

```

sources cc-cli-env.sh and injects

Flink SQL reports – runtime split

CCAF=7 reports/Protobuf+SR),

CP Flink SQL: 00_source_table,

05_isotope_view,

05_report_sinks (avro-confluent),
10/20/25/30/40/60/70 INSERT INTO

(CCAF SQL is inlined under

  optional AI extension (CCAF-only,

PoC: 8th, AI-generated report –

stuck-trace alert into an LLM root-

hypothesis via CREATE MODEL +

(wired in by terraform/setup-ccaf-

CCAF infrastructure-as-code (`make

Confluent provider – cloud

required Terraform (>= 1.13) +

confluent_api_key/secret, cloud,

organization lookup + other data

environment (ESSENTIALS stream-

Kafka cluster + Kafka API key

(iac-confluent-api_key_rotation-

service account + 6 role bindings,

artifact upload, SR API key

`confluent_flink_statement`

+ 3 VIEW + 7 sink CREATE TABLE + 2

2 CREATE FUNCTION (both PTFs) + 7

OPTIONAL AI trace-RCA report – 3

```

```

sink +

false)
  outputs.tf
rotating

(sensitive)
docs/
from the README)
  design.md
deep-dive)
  runbook-minikube.md
4.4)
  runbook-ccaf.md
($ 4.5)
  metrics.md
PromQL reference ($ 4.6)
  terraform.png
in $ 4.5)
Makefile
flink-reports-up /

reports-down /

metrics-delete / ...

statements (CREATE MODEL + Protobuf

INSERT ... ML_PREDICT); gated on
var.enable_trace_rca (default

environment_id, bootstrap, SR URL,

Kafka + SR API key/secret outputs

extracted long-form docs (linked

how the isotope is carried ($ 1.0

full CP-on-Minikube run sequence ($

full CCAF / Terraform run sequence

Micrometer/Prometheus meter +

rendered resource graph (embedded

cp-up / flink-up / kafka-pf-up /

cc-flink-reports-up / cc-flink-

metrics-up / metrics-down /

```

4.0 Running

Cheapest-first order if you're new: `./gradlew test` (§ 4.1) → local CP via Minikube (§ 4.2–4.4) → CCAF in the cloud (§ 4.5). Skip ahead if you only care about one runtime.

Just want the commands? The full local stack-up sequence (cluster → Kafka → Flink → reports → traffic → teardown) is consolidated in [docs/runbook-minikube.md](#).

4.1. Unit tests (no broker, instant)

```

./gradlew test                                # runs :ptf:test (the app has no unit
tests in this repo)
# or scoped:
./gradlew :ptf:test                            # TDigestsTest – T-Digest sketch
(de)serialization + accuracy

```

Note: the isotope codec / consume-marker unit tests (`IsotopeCodecTest`, `IsotopeContextRecordConsumeTest`) now live in the standalone tracing library `j3-signalroom/kafka-isotope`, which this repo consumes as the `ai.signalroom:kafka-isotope-core` / `kafka-isotope-metrics` dependencies. The `:app` subproject here keeps only integration tests (§ 4.3).

4.2 Demo CLI — see one trace propagate live

The fastest way to watch the isotope mechanic. Requires the cluster to be up and the Kafka + SR forwards running (see step 3 below for the bring-up commands). The CLI has two argument styles — **pipeline-position verbs** that bake in the orders.* topic chain (recommended for the demo) and **generic verbs** that take raw topic + service args (for ad-hoc inspection on any topic):

Verb	Args	What it does
place	[payload]	Produces one isotope-tagged DemoEvent to orders.placed as order-intake-service (default payload: hello), then exits. Auto-creates the topic.
enrich	—	Consumes from orders.placed , adopts the isotope, emits a consume-edge marker to isotope_consume_edge_markers as order-enrichment-service , then re-produces to orders.enriched . Runs until Ctrl-C.
fulfill	—	Same as enrich but for orders.enriched → orders.fulfilled as order-fulfillment-service . Runs until Ctrl-C.
ship	—	Terminal consumer for orders.fulfilled as shipping-notification-service . Emits a consume-edge marker so it shows up in the bipartite report; does not re-produce. Runs until Ctrl-C.
send	<topic> <service> <payload>	Generic produce. Auto-creates the topic.
hop	<in-topic> <out-topic> <service>	Generic consume-then-produce; emits a consume-edge marker. Runs until Ctrl-C.
consume	<topic> <service>	Generic terminal-consume; emits a consume-edge marker and pretty-prints the trail. Runs until Ctrl-C.
sink	<topic>	Passive peek — pretty-prints the isotope trail but does NOT emit a consume marker. Use for ad-hoc inspection. Runs until Ctrl-C.

4.2.1 Full Bipartite Demo

A 4-stage chain (full bipartite graph) in four terminals. Run them in pipeline order — **place** produces, then **enrich** / **fulfill** / **ship** each pick up where the previous stage left off:

```
# Terminal A — kick the chain off (run repeatedly to send more)
./gradlew :app:run --args="place 'hello world'" -q

# Terminal B — first hop: orders.placed → orders.enriched
./gradlew :app:run --args="enrich" -q
```

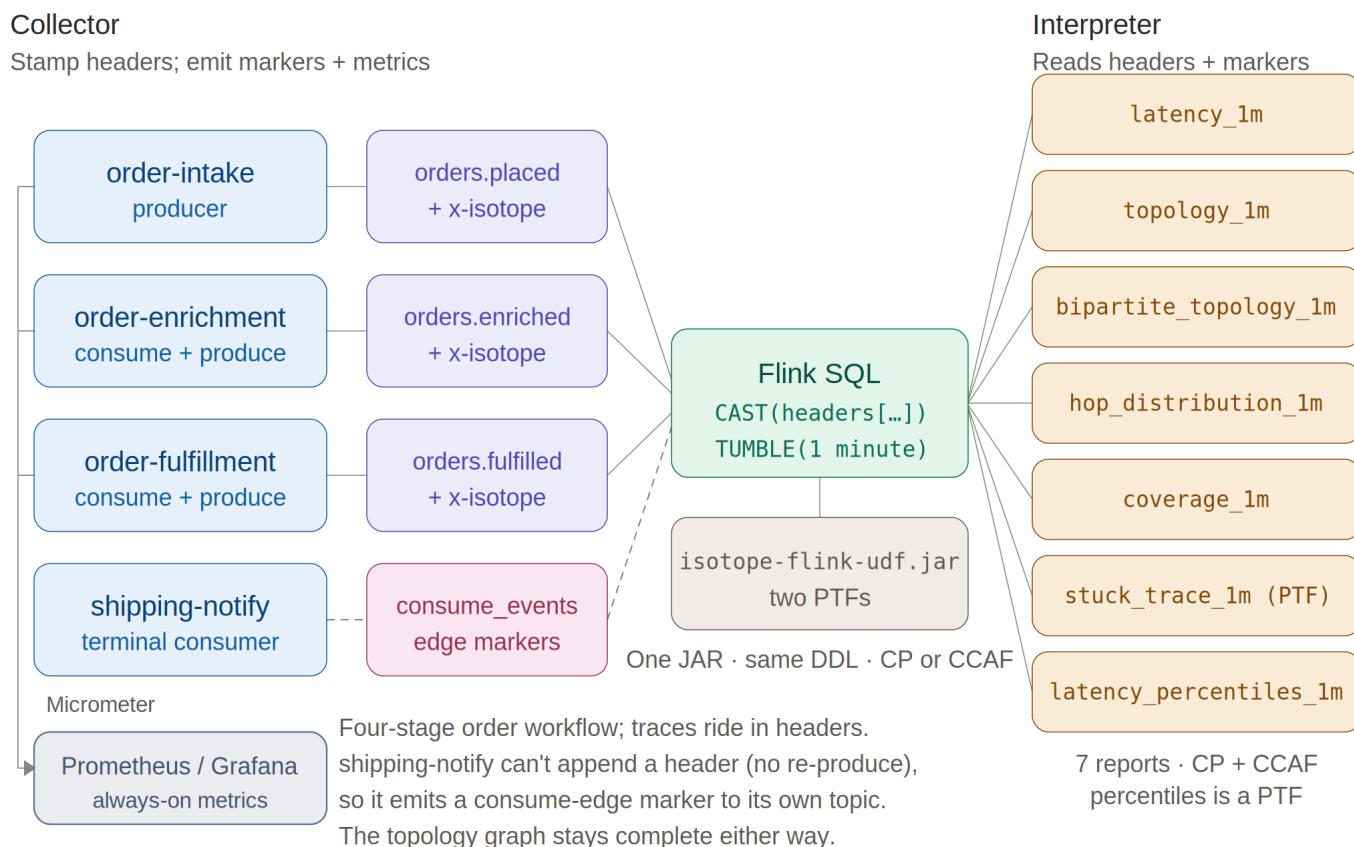
```
# Terminal C – middle hop: orders.enriched → orders.fulfilled
./gradlew :app:run --args="fulfill" -q

# Terminal D – terminal consumer (prints the full 3-hop trail AND emits a
#                 consume-edge marker so shipping-notification-service shows
up
#                 in the bipartite report)
./gradlew :app:run --args="ship" -q
```

Terminal D's output for each record shows the same `trace_id` across all three hops, `origin = order-intake-service` (never reassigned), and `hops[]` listing `order-intake-service → order-enrichment-service → order-fulfillment-service` in order with per-hop timestamps. The bipartite-topology report sees all six edges: produce edges `order-intake-service → orders.placed`, `order-enrichment-service → orders.enriched`, `order-fulfillment-service → orders.fulfilled` and consume edges `orders.placed → order-enrichment-service`, `orders.enriched → order-fulfillment-service`, `orders.fulfilled → shipping-notification-service`. Swap `consume` for `sink` if you only want to inspect records without recording the terminal edge. Override endpoints via `-Dkafka.bootstrap=...` / `-Dschema.registry.url=...` if you're not on the default Minikube layout.

Collector

Stamp headers; emit markers + metrics



4.3 Integration tests (live Kafka via Minikube)

Bring up the local Confluent Platform stack and port-forward Kafka + SR:

```
make minikube-start          # one-time
make cp-up                   # CFK Operator +
Kafka/SR/Connect/ksqlDB/C3 (~5 min)
make kafka-pf-up             # localhost:30092 → Kafka,
                              localhost:8081 → Schema Registry
```

Then run the suite:

```
./gradlew :app:integrationTest #
every IT below
./gradlew :app:integrationTest --tests '*ProducerInterceptorIT' #
just one
```

Override the endpoints if needed:

```
./gradlew :app:integrationTest \
  -PkafkaBootstrap=localhost:30092 \
  -PschemaRegistryUrl=http://localhost:8081
```

Tear down forwards when done:

```
make kafka-pf-down
```

The integration tests cover:

Test	What it verifies
BrokerSmokeIT	AdminClient can create/list/delete a topic via the NodePort port-forward
ProducerInterceptorIT	A consumer sees the <code>x-isotope</code> JSON header + all 7 scalar reporting headers with the expected origin/hop values, and the Protobuf round-trip preserves <code>DemoEvent.source</code> / <code>payload</code>
ThreeStageHopPropagationIT	<code>order-intake-service</code> → <code>topic-AB</code> → <code>order-enrichment-service</code> → <code>topic-BC</code> → <code>order-fulfillment-service</code> produces a stable trace ID, 2-hop trail in send order, and correct scalar headers (origin = <code>order-intake-service</code> , this = <code>order-enrichment-service</code> , hop count = 2) at the terminal; consume-then-produce hops use <code>IsotopeContext.adoptFromRecord</code> to carry the trace forward

Test	What it verifies
BipartiteTopologyIT	The 4-stage <code>order-intake-service</code> → <code>topic-AB</code> → <code>order-enrichment-service</code> → <code>topic-BC</code> → <code>order-fulfillment-service</code> → <code>topic-CD</code> → <code>shipping-notification-service</code> chain emits exactly three consume-edge markers to a per-test markers topic — one per consume edge. Every marker carries the trace ID, forwarded <code>x-isotope-*</code> scalars describing the upstream producer, and the new <code>x-isotope-consumer-service</code> naming the downstream consumer. Asserts the <code>(consumer_service, consumed_topic)</code> set is exactly the three pairs of stages 2-4

4.4 Flink SQL reports on Confluent Platform for Apache Flink (Minikube)

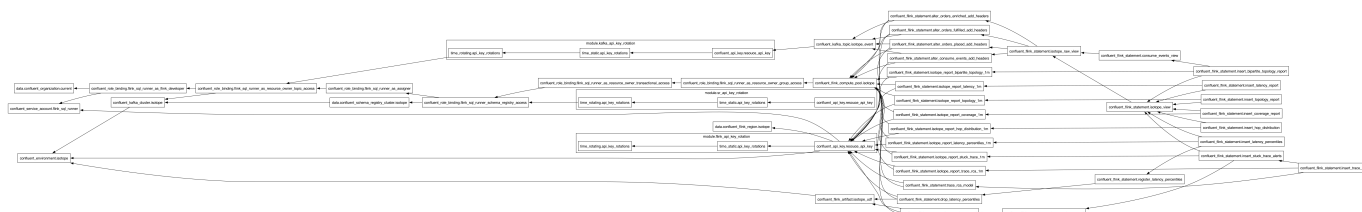
Seven reports — five pure Flink SQL plus two JAR-backed PTFs — run against a `cp-flink` session cluster (Flink 2.1.2) managed by the Confluent Flink Kubernetes Operator. The same FQL files deploy to Confluent Cloud — see § 4.5 for that path; this section is the local-Minikube one.

The full bring-up sequence — `cluster` → `Flink` → `reports` → `traffic` → `teardown` — is consolidated in [docs/runbook-minikube.md](#). The short version: `make flink-up` then `make flink-reports-up`, then drive traffic across **multiple** 1-minute windows (a single burst sits in one open window forever — the watermark has to cross `window_end` for a tumbling window to emit) and wait ~90s after the last record.

Report sink topics ride **Avro+SR** (`avro-confluent`, auto-registered on first write) so Control Center renders them natively — a deliberate *format-by-domain* split: app events are **Protobuf+SR** (`DemoEvent`), Flink aggregates are **Avro+SR** (cp-flink ships no SR-integrated Protobuf format), and the consume-edge marker topic `isotope_consume_edge_markers` is **null-value / headers-only**. Not a defect — a clean split by domain.

4.5 Flink SQL reports on Confluent Cloud for Apache Flink (CCAF)

The CCAF parallel of § 4.4, driven by Terraform under [terraform/](#) — no local cluster. One `make` target spins up a fresh Confluent Cloud environment, Kafka cluster, compute pool, two rotating service-account API key pairs (Kafka + Schema Registry), the uploaded PTF JAR, and 25 `confluent_flink_statement` resources (4 ALTER TABLE + 3 VIEW + 7 sink CREATE TABLE + 2 CREATE FUNCTION + 7 INSERT INTO, plus 2 transient DROP FUNCTION). The shape mirrors `apache_flink-kickstarter-ii` — same provider version and `iac-confluent-api_key_rotation-tf_module`.



The full `provision` → `deploy` → `traffic` → `teardown` sequence is consolidated in [docs/runbook-ccaf.md](#). The short version: `export` a Cloud-level Confluent Cloud API key, `make cc-flink-reports-up` (~6–8 min first run; idempotent re-applies), drive traffic with `scripts/cc-app-run.sh place|enrich|fulfill|ship` across **multiple** 1-minute windows (a single burst sits in one open

window forever, and you wait ~90s after the last record), then `make cc-flink-reports-down` to `terraform destroy` the whole environment.

Two CCAF-specific design points worth keeping in the README:

Format-by-runtime (not -by-domain). CP's reports land on **Avro+SR** (`'value.format' = 'avro-confluent'` in [scripts/flink/sql/cp/05_report_sinks.fql](#)). CCAF's reports land on **Protobuf+SR** (`'value.format' = 'proto-registry'` in each sink's WITH clause in [terraform/setup-confluent-flink.tf](#)). The two runtimes' SQL is otherwise unshared: CP's lives hardcoded in [scripts/flink/sql/cp/](#), CCAF's lives inline as `confluent_flink_statement` resources in [terraform/setup-confluent-flink.tf](#).

Why percentiles is a PTF. CCAF rejects all `CREATE FUNCTION` statements for user-defined *aggregate* functions ("aggregate functions are not supported"). Percentiles would naturally be an aggregate, so to keep the report portable it's implemented as a `ProcessTableFunction` instead — `LATENCY_PERCENTILES` (class `LatencyPercentilesPTF`) does its own 1-minute tumbling-window aggregation over a T-Digest sketch via per-window state and event-time timers. A PTF has no such restriction, so it registers and runs on **both** runtimes, exactly like `STUCK_TRACE_PTF`. Both runtimes therefore run the same seven reports: `latency` (avg/min/max), `topology` (produce-side), `bipartite_topology` (full service↔topic↔service graph), `hop_distribution`, `coverage`, `stuck_trace`, and `latency_percentiles` (p50/p95/p99).

Optional: AI root-cause analysis (CCAF-only, off by default). Beyond the seven deterministic reports, [terraform/setup-ccaf-ai.tf](#) wires an **eighth, AI-generated report** that turns each stuck-trace *alert* into a natural-language root-cause hypothesis plus a one-line remediation. It adds three `confluent_flink_statement` resources — `CREATE MODEL trace_rca` (a remote text-generation model), a `proto-registry` sink `isotope_report_trace_rca_1m`, and an `INSERT ... SELECT ... LATERAL TABLE(ML_PREDICT('trace_rca', ...))` — all **gated on `var.enable_trace_rca` (default `false`)**, so a normal `make cc-flink-reports-up` is completely unaffected. The model is invoked once per *alert* (the low-volume 1-minute stuck-trace report topic), never per record, and its output lands on its own topic — it never overwrites the deterministic reports. Defaults target OpenAI (`gpt-4o`); for Claude hosted by Anthropic — a **direct** CCAF provider — set `rca_model_provider = "anthropic"`, `rca_model_endpoint = "https://api.anthropic.com/v1/messages"`, a Claude `rca_model_version`, and your Claude `rca_model_api_key` (bare API key, no AWS auth), plus the Anthropic-required `rca_model_max_tokens`. Claude via AWS Bedrock (`rca_model_provider = "bedrock"`) also works but isn't required. The standalone SQL PoC — a `CREATE MODEL + ML_PREDICT` walkthrough with provider notes and alternative options — is in [scripts/flink/sql/ccaf-ai/trace_rca.fql](#).

4.6 Stateless reports via Micrometer → Prometheus/Grafana (optional)

Three of the seven reports — `latency_1m`, `topology_1m`, `hop_distribution_1m` — are pure stateless scalar aggregation over bounded-cardinality dimensions (service / topic / hop_count, never `trace_id`). Those don't need a stream processor: the `producer interceptor` already has every value in scope on each `send()`, so it emits them as **Micrometer** meters (`PrometheusIsotopeMetrics.java`) and lets **Prometheus** window them at read time, with **Grafana** on top. The other four (`latency_percentiles`, `coverage`, `bipartite_topology`, `stuck_trace`) are per-`trace_id` stateful or absence-of-event problems Prometheus can't express, so they **stay in Flink**. This path is **additive and opt-in** — a 3-Micrometer / 4-Flink split.

Full details — the meter/PromQL reference, the produce- and consume-side signals, the two deliberate gaps (`distinct_traces`, windowed `min`), and why percentiles stays a PTF — are in [docs/metrics.md](#). The one-command Prometheus + Grafana showcase has its own runbook: [k8s/monitoring/README.md](#) (`make metrics-up`).

5.0 Resources

- [Medium Article: Kafka's quiet observability superpower — Kafka Interceptors](#)